

PROBING WHAT MUSIC CLASSIFIERS ACTUALLY LEARN: A MULTI-METHOD EXPLAINABILITY ANALYSIS OF GENRE CLASSIFICATION ON GTZAN

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ABSTRACT

Convolutional neural networks (CNNs) achieve competitive accuracy on music genre classification, yet it remains unclear whether their decisions are grounded in musicologically meaningful features or in dataset artifacts such as recording noise, silence patterns, and artist fingerprints. This paper presents a systematic, multi-method explainability analysis of a CNN trained on the GTZAN dataset, employing three complementary techniques: SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and attention maps from a fine-tuned Audio Spectrogram Transformer (AST). A quantitative validation framework is introduced comprising four metrics: musicological alignment, faithfulness, inter-method agreement, and spurious correlation detection, to rigorously evaluate explanation quality. The results reveal a notable genre-dependent dichotomy: some genres are classified based on genuinely musicological spectral features, while others are driven by recording artifacts and silence-region characteristics. Moreover, SHAP and LIME produce consistently low-agreement explanations for the same predictions, demonstrating that explanation narratives are strongly method-dependent. These findings underscore the necessity of multi-method triangulation and domain-grounded validation when deploying explainability tools in music information retrieval.

1. INTRODUCTION

Music genre classification has served as a foundational benchmark in music information retrieval (MIR) since the seminal work of Tzanetakis and Cook [1]. Over the past decade, deep learning models operating on mel spectrogram representations have driven classification accuracy on the GTZAN benchmark to near-perfect levels. Recent architectures, including capsule neural networks with aggressive data augmentation, have reported accuracies as high as 99.9% [2], and other deep architectures have independently exceeded 99% [3], suggesting that the problem

is approaching saturation on this benchmark. In terms of raw classification performance, there is little room left to gain.

Yet this apparent ceiling of discovery masks a more fundamental and largely unresolved question: what are these models actually learning? High classification accuracy on GTZAN does not necessarily indicate that a model has internalised meaningful musical characteristics such as rhythm, harmony, or timbre. Prior work has demonstrated that classifiers may instead exploit dataset-specific artifacts, including recording conditions, duplicate clips, and labelling inconsistencies, to achieve high scores while failing to capture genuine genre semantics [4, 5]. A model that achieves 99% accuracy by recognising microphone characteristics is fundamentally different from one that has learned the 12-bar blues structure or the sub-bass patterns of hip-hop, even though their test metrics may be identical. The research frontier, therefore, has shifted from improving accuracy to interrogating the nature of what is learned.

This distinction is not merely academic. In MIR, models are increasingly deployed in production systems for recommendation, cataloguing, and rights management [6]. If these systems base decisions on spurious correlations (microphone characteristics, compression artifacts, or silence padding), they will fail unpredictably on data that does not share these incidental properties. The field of explainable Artificial Intelligence (XAI) [7] offers tools to interrogate model decisions, yet XAI methods have seen limited rigorous application in MIR.

Prior work on explainability in music classification has been sparse and methodologically narrow. Mishra et al. [8] applied LIME to music content analysis but did not validate whether the identified regions corresponded to musically meaningful features. Haunschmid et al. [9] proposed audioLIME with source-separation-based explanations, advancing interpretability but not systematically evaluating faithfulness or cross-method consistency. No existing study, to the best of current knowledge, applies multiple complementary XAI techniques to the same model and validates the resulting explanations against both musicological ground truth and quantitative faithfulness metrics.

The GTZAN dataset [1] presents a particularly instructive test case. Despite being the most widely used benchmark in genre classification, GTZAN contains well-documented flaws: artist repetition within genres, incon-



sistent recording quality across genres, and faulty or mis-
labelled clips [5]. These flaws are not merely a limitation;
they serve as a natural instrument for probing whether clas-
sifiers learn music or metadata. As Sturm [10] argued, a
genre classifier may function as a “horse”, a system that
exploits dataset regularities without learning the intended
concept.

This paper addresses the following research questions:

RQ1: To what extent do CNN-based genre classifiers at-
tend to musicologically meaningful spectral features
versus dataset artifacts?

RQ2: Do different XAI methods (SHAP, LIME, attention)
provide consistent explanations for the same model
decisions?

RQ3: Can genuine musicological learning be quantita-
tively distinguished from spurious correlation ex-
ploitation?

The contributions of this paper are as follows: (1)
a multi-method explainability pipeline applying SHAP,
LIME, and Transformer attention to the same CNN and
dataset; (2) a four-metric quantitative validation frame-
work for explanation quality; (3) empirical evidence that
genre classifiers exhibit a genre-dependent spectrum from
genuine musicological learning to artifact exploitation; and
(4) a demonstration that XAI method choice substantially
affects the explanation narrative, arguing for multi-method
triangulation as standard practice.

2. RELATED WORK

2.1 Music genre classification

Music genre classification has been a central task in mu-
sic information retrieval (MIR) since the pioneering work
of Tzanetakis and Cook [1], which introduced a feature-
based approach using timbral texture, rhythmic content,
and pitch features. Early systems relied on handcrafted de-
scriptors such as MFCCs, spectral centroid, zero-crossing
rate, and beat histograms, combined with classical clas-
sifiers including k-nearest neighbours and support vector
machines [?].

The field has since shifted toward deep learning ap-
proaches that learn representations directly from au-
dio. Convolutional neural networks (CNNs) operating on
mel spectrograms have become the dominant paradigm,
demonstrating strong performance across multiple MIR
tasks [11, 12]. Variants incorporating residual connec-
tions, attention mechanisms, and multi-scale feature ex-
traction have further improved performance [13]. More re-
cently, Transformer-based architectures such as the Audio
Spectrogram Transformer (AST) [14] and large-scale self-
supervised models like MERT [15] have pushed perfor-
mance closer to saturation by leveraging large pre-training
corpora.

Despite these advances, performance gains on GTZAN
have plateaued, with multiple studies reporting accuracies
exceeding 95% and, in some cases, approaching 100%.

This saturation raises questions about whether further im-
provements reflect genuine advances in musical under-
standing or exploitation of dataset-specific regularities.

2.2 GTZAN dataset critique and shortcut learning

The GTZAN dataset [1], while widely used, has been
extensively criticised for its structural flaws. Sturm [5]
conducted a comprehensive audit, identifying duplicate
recordings, mislabelled tracks, and substantial artist over-
lap between training and test sets. These issues introduce
confounding factors that allow models to achieve artifi-
cially high performance without learning meaningful genre
characteristics.

This phenomenon is closely related to the concept of
shortcut learning, where models exploit spurious correla-
tions that are predictive in the training data but not causally
related to the task [?]. In the context of GTZAN, such
shortcuts may include recording conditions, production
styles, or artist-specific characteristics rather than genre-
defining musical features. Sturm [10] characterised this
issue through the “horse” analogy, highlighting that clas-
sifiers may appear successful while relying on unintended
cues.

Subsequent work has reinforced these concerns by
demonstrating that simple baselines exploiting metadata
or recording artifacts can achieve competitive performance
[?]. More broadly, the problem of dataset bias and spuri-
ous correlations has been widely studied in machine learn-
ing, with implications for model robustness and generali-
sation [?]. These findings motivate the need for methods
that go beyond accuracy to interrogate what models actu-
ally learn.

2.3 Explainability in MIR

Explainable Artificial Intelligence (XAI) has emerged as
a critical tool for understanding model behaviour, particu-
larly in domains where interpretability and trust are essen-
tial [7]. In MIR, however, the application of XAI remains
relatively limited compared to computer vision and natural
language processing.

Early work by Mishra et al. [8] applied LIME [16] to
music content analysis, demonstrating that perturbation-
based explanations can highlight salient spectrogram re-
gions. Hauns Schmid et al. [9] proposed audioLIME, which
improves upon standard LIME by generating perturbations
through source separation, yielding more musically mean-
ingful explanations. Gradient-based attribution methods,
including saliency maps and Grad-CAM [17], have also
been applied to audio classification tasks, though often
without rigorous validation against domain knowledge.

More recently, attention-based models such as AST
[14] have been used to provide interpretability through at-
tention visualisation. However, the extent to which atten-
tion weights correspond to causal feature importance re-
mains debated [18]. This has led to increasing recognition
that no single explanation method is sufficient to fully char-
acterise model behaviour.

2.4 Evaluating explanations and the disagreement problem

A central challenge in XAI is the evaluation of explanation quality. Metrics such as faithfulness, sensitivity, and robustness have been proposed to assess whether explanations accurately reflect model decision processes [19, 20]. However, these metrics are rarely applied in MIR, where qualitative inspection of saliency maps remains common practice.

Another critical issue is the *disagreement problem*: different XAI methods often produce substantially different explanations for the same prediction [21]. This has been observed across domains and raises concerns about the reliability of individual explanation techniques. In audio and MIR contexts, where perceptual structure is complex and multi-dimensional, this problem is likely to be even more pronounced.

Despite these challenges, there has been no systematic study in MIR that (1) applies multiple complementary XAI methods to the same model, (2) quantitatively evaluates explanation quality, and (3) grounds explanations in domain-specific (musicological) expectations. This paper addresses this gap by introducing a multi-method, quantitatively validated framework for analysing what genre classifiers learn on GTZAN.

3. METHODOLOGY

3.1 Dataset and pre-processing

The GTZAN dataset [1] is used, comprising 1,000 audio clips of 30 seconds each, distributed equally across 10 genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. Following the audit methodology of Sturm [5], known faulty clips (exact duplicates, severely distorted recordings) are flagged and excluded, restricting the analysis to clean clips.

Audio is loaded at 22,050 Hz and converted to 128-bin mel spectrograms using a 2,048-sample FFT window with 512-sample hop length, computed via librosa [22]. For the CNN, 3-second segments are randomly cropped from each 30-second clip during training (providing implicit data augmentation), converted to log-mel scale, and resized to 128×128 pixels. For the AST, 10-second segments are resampled to 16,000 Hz to match the pre-trained model’s expected input format.

The dataset is split into 80% training and 20% test sets using stratified sampling to ensure balanced genre representation, yielding approximately 800 training and 200 test clips.

3.2 CNN architecture (MusicCNN)

The subject model is a purpose-built CNN designed for interpretability research. The architecture consists of three convolutional blocks, each containing two 3×3 convolutional layers with batch normalisation [23], ReLU activation, 2×2 max pooling, and spatial dropout (rate 0.15). Channel dimensions progress as $1 \rightarrow 32 \rightarrow 64 \rightarrow 128$.

An adaptive average pooling layer produces 4×4 spatial features, followed by a two-layer fully connected head ($2,048 \rightarrow 256 \rightarrow 10$) with dropout (rate 0.4). The model totals approximately 814K parameters.

Residual connections and complex skip pathways are deliberately avoided to ensure that feature attribution methods can trace contributions through the network without ambiguity. This design choice prioritises interpretability over raw performance.

Training employs the AdamW optimiser [24] with an initial learning rate of 10^{-3} , cosine annealing schedule with linear warmup [25], and cross-entropy loss with label smoothing of 0.1. Data augmentation includes SpecAugment [26] (frequency and time masking) and mixup [27] ($\alpha = 0.3$). Training proceeds for 120 epochs with a batch size of 32. All experiments use PyTorch [28] on Apple MPS (Metal Performance Shaders) hardware acceleration.

3.3 Explainability methods

Three complementary XAI techniques are applied, each grounded in different theoretical assumptions:

3.3.1 SHAP (SHapley Additive exPlanations)

SHAP [29] is rooted in cooperative game theory, assigning each input feature an importance value based on Shapley values [30]. The DeepExplainer variant is employed, using a background distribution of 50 stratified training spectrograms (5 per genre) to compute expected model behaviour. For each correctly classified test clip, SHAP produces a 128×128 attribution map indicating how each mel spectrogram bin contributed to the predicted class.

SHAP offers theoretical guarantees of local accuracy, missingness, and consistency that other attribution methods lack. However, the gradient-based approximation in DeepExplainer introduces some approximation error, and the method operates at the individual pixel level without awareness of perceptual grouping.

3.3.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME [16] treats the model as a black box, perturbing the input and observing output changes to fit a local interpretable surrogate model. Each mel spectrogram is segmented into superpixels using the SLIC algorithm, 1,000 perturbations are generated by randomly masking subsets of superpixels, and a weighted Ridge regression model is fitted to approximate the local decision boundary.

Unlike SHAP, LIME operates at the superpixel level, producing coarser but potentially more perceptually meaningful attributions. Its model-agnostic nature means it can be applied to any classifier without access to gradients.

3.3.3 AST attention maps

As a third perspective, an Audio Spectrogram Transformer (AST) [14] pre-trained on AudioSet is fine-tuned and CLS token attention weights are extracted from the final self-attention layer. The AST processes each audio clip as a sequence of overlapping 16×16 spectrogram patches,

and the CLS token’s attention distribution indicates which patches the model attends to when making its classification decision.

A two-phase fine-tuning strategy is employed: 10 epochs of linear probing (frozen backbone, trained head only) followed by 20 epochs of full fine-tuning with a reduced learning rate (10^{-5}). Cross-entropy loss with label smoothing of 0.1 is used throughout.

The known caveat that attention weights do not necessarily reflect causal importance [18] is acknowledged. Nevertheless, attention provides a complementary signal, especially valuable when compared against gradient-based (SHAP) and perturbation-based (LIME) methods.

3.4 Quantitative validation framework

A central contribution of this work is a four-metric framework for evaluating explanation quality in the context of music classification:

3.4.1 Musicological alignment (ρ_{align})

Expected frequency band importance rankings are defined for each genre based on established music theory. Five canonical frequency bands are considered: sub-bass (0–60 Hz), bass (60–250 Hz), mid (250–2,000 Hz), upper-mid (2,000–6,000 Hz), and brilliance (6,000–20,000 Hz). For each genre, a musicological ranking is assigned (e.g., hip-hop is expected to show highest attribution in sub-bass/bass; classical in upper-mid/brilliance).

For each genre, the mean absolute SHAP value within each frequency band is computed, these bands are ranked by attribution magnitude, and the Spearman rank correlation (ρ) between the observed and expected rankings is computed:

$$\rho_{align} = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (1)$$

where d_i is the rank difference for band i and $n = 5$ is the number of frequency bands. Values near +1 indicate that the model attends to musicologically expected frequency regions; values near −1 suggest reliance on unexpected or artifactual features.

3.4.2 Faithfulness

Faithfulness measures whether the identified important regions are genuinely used by the model. We follow the occlusion protocol of Samek et al. [19] and Hooker et al. [20]: for each explained clip, we progressively mask the top- $k\%$ of pixels (by SHAP magnitude) with zeros and measure the drop in prediction confidence:

$$\text{Faith}(k) = P(\hat{y} \mid \mathbf{x}) - P(\hat{y} \mid \mathbf{x}_{\setminus \text{top-}k}) \quad (2)$$

where $P(\hat{y} \mid \mathbf{x})$ is the original confidence and $P(\hat{y} \mid \mathbf{x}_{\setminus \text{top-}k})$ is the confidence after masking. A large drop confirms that SHAP correctly identified decision-critical regions. We evaluate at $k \in \{5, 10, 20, 30, 50\}\%$.

Genre	Correct/Total	Accuracy (%)	Level
Classical	20/20	100.0	Excellent
Jazz	19/19	100.0	Excellent
Blues	19/20	95.0	Excellent
Hip-hop	19/20	95.0	Excellent
Metal	17/20	85.0	Good
Pop	17/20	85.0	Good
Country	15/20	75.0	Moderate
Reggae	14/20	70.0	Moderate
Disco	13/20	65.0	Moderate
Rock	13/20	65.0	Moderate
Overall	166/199	83.4	

Table 1. Per-genre test accuracy of the MusicCNN on GTZAN. Genres are ordered by accuracy.

3.4.3 Inter-method agreement

To quantify the disagreement problem [21], we compute three agreement metrics between SHAP and LIME attribution maps: Pearson correlation coefficient r , Spearman rank correlation ρ , and intersection-over-union (IoU) of the top-10% most important pixels:

$$\text{IoU}_{10} = \frac{|S_{10}^{\text{SHAP}} \cap S_{10}^{\text{LIME}}|}{|S_{10}^{\text{SHAP}} \cup S_{10}^{\text{LIME}}|} \quad (3)$$

3.4.4 Spurious correlation detection

We compare SHAP attributions in spectrogram regions likely to contain artifacts versus regions containing musical content. Two ratios are computed: (1) the silence ratio, comparing mean SHAP magnitude in low-energy (silence) bins to high-energy bins; and (2) the edge ratio, comparing SHAP in temporal edge regions (first/last 10% of the spectrogram) to centre regions. A silence ratio exceeding 1.0 indicates the model assigns more importance to silence than to musical content, which is a strong indicator of artifact exploitation.

4. RESULTS

4.1 Classification performance

The MusicCNN achieves an overall test accuracy of 83.4% on 199 test clips, with substantial per-genre variation (Table 1). Classical and jazz are classified perfectly (100%), while disco and rock prove most challenging (65%). These confusion patterns align with musicological similarity: disco and rock share similar rhythmic drive and spectral energy distributions, while classical and jazz have highly distinctive timbral profiles.

Figure 1 displays the full confusion matrix. The primary confusion pairs are disco↔rock (4 mutual misclassifications), country↔rock (3), and reggae↔country (2). These confusions provide meaningful targets for XAI interrogation: understanding *why* the model confuses these pairs may reveal whether it has learned the relevant distinguishing features.

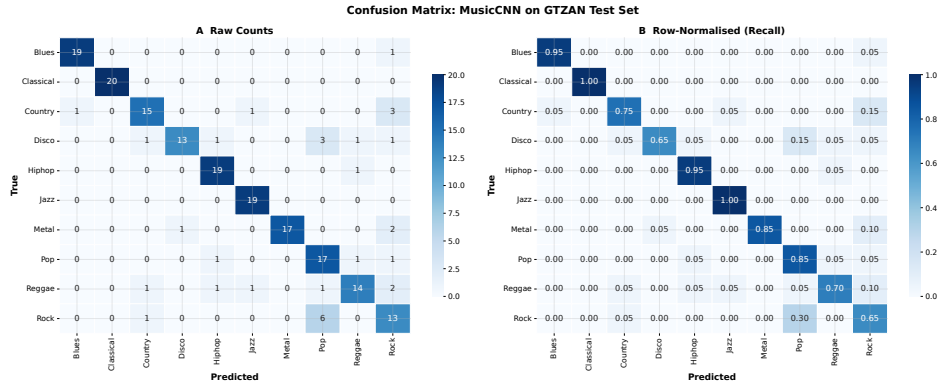


Figure 1. Confusion matrix for MusicCNN on the GTZAN test set. Classical and jazz show perfect separation; disco and rock exhibit the highest mutual confusion.

4.2 SHAP explanations

Figure 2 presents SHAP attribution overlays for representative correctly classified clips from each genre. The overlays reveal qualitatively distinct attribution patterns across genres. Classical clips show distributed high-frequency attribution consistent with orchestral overtone richness. Hip-hop shows concentrated low-frequency attribution aligned with bass-heavy production. Jazz exhibits broad spectral attribution reflecting harmonic complexity.

Figure 3 aggregates SHAP attributions by frequency band across genres. Notable patterns include the dominance of low-frequency bands for hip-hop and reggae, mid-range salience for country and rock, and high-frequency emphasis for classical and metal.

4.3 LIME explanations

Figure 4 shows LIME superpixel explanations for the same clips. LIME identifies broader contiguous regions compared to SHAP’s pixel-level attributions, reflecting the superpixel granularity of the perturbation scheme.

Figure 5 provides a consolidated view of all four quantitative XAI metrics discussed in the following subsections.

4.4 Faithfulness evaluation

Figure 6 presents faithfulness curves for all genres. Masking the top 5% of SHAP-identified pixels produces confidence drops ranging from 29% (hip-hop) to 72% (classical), confirming that SHAP identifies genuinely decision-critical regions. At the 50% masking level, all genres show substantial confidence degradation (Table 2).

Disco (0.917) and pop (0.887) exhibit the highest faithfulness, indicating highly localised decision strategies. Reggae (0.380) shows the lowest faithfulness, suggesting the model relies on diffuse, distributed spectral cues rather than localised features for this genre.

4.5 Musicological alignment

Table 3 and Figure 7 present the Spearman correlation between observed SHAP band rankings and musicological

Genre	Original Conf.	Faith. ($k = 50\%$)
Blues	0.958	0.741
Classical	0.807	0.626
Country	0.903	0.817
Disco	0.984	0.917
Hip-hop	0.924	0.819
Jazz	0.489	0.453
Metal	0.584	0.530
Pop	0.914	0.887
Reggae	0.665	0.380
Rock	0.600	0.472

Table 2. Faithfulness at $k = 50\%$ masking: the confidence drop when removing the top half of SHAP-attributed pixels. Higher values indicate more faithful explanations.

expectations. The results reveal a striking genre-dependent pattern:

Hip-hop shows near-perfect positive alignment ($\rho = 0.95$, $p = 0.051$): the model attends primarily to sub-bass and bass regions, precisely where hip-hop’s 808-style bass and rhythmic percussion reside. This constitutes strong evidence that the CNN has learned a genuine musicological feature for this genre.

Country ($\rho = -1.0$, $p < 0.001$) and **pop** ($\rho = -0.95$, $p = 0.051$) exhibit near-perfect *inverse* alignment. The model attends to frequency bands that are musicologically *least* expected to be discriminative for these genres, strongly suggesting reliance on recording artifacts, noise floor characteristics, or other non-musical features.

The remaining genres fall along a spectrum between these extremes, with reggae showing weak positive alignment ($\rho = 0.20$) and blues and rock showing substantial inverse tendencies ($\rho = -0.80$).

4.6 Inter-method agreement

Table 4 and Figure 8 quantify the agreement between SHAP and LIME attributions. Across all genres, agreement is remarkably low: Pearson correlations range from $r = -0.01$ (disco) to $r = 0.20$ (metal), and top-10% IoU

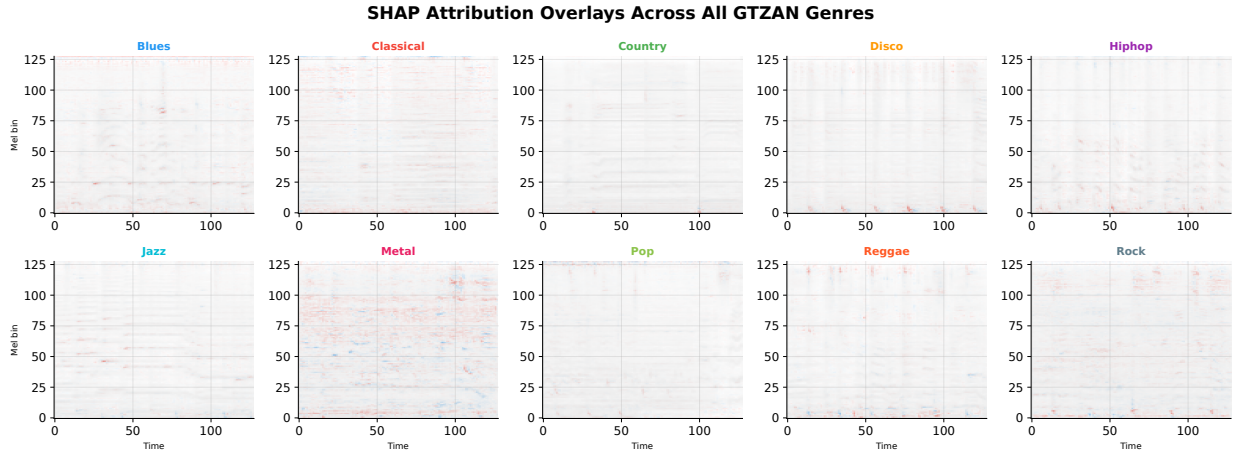


Figure 2. SHAP attribution overlays on mel spectrograms for correctly classified clips from each genre. Red indicates positive attribution (pushes toward the predicted class); blue indicates negative attribution.

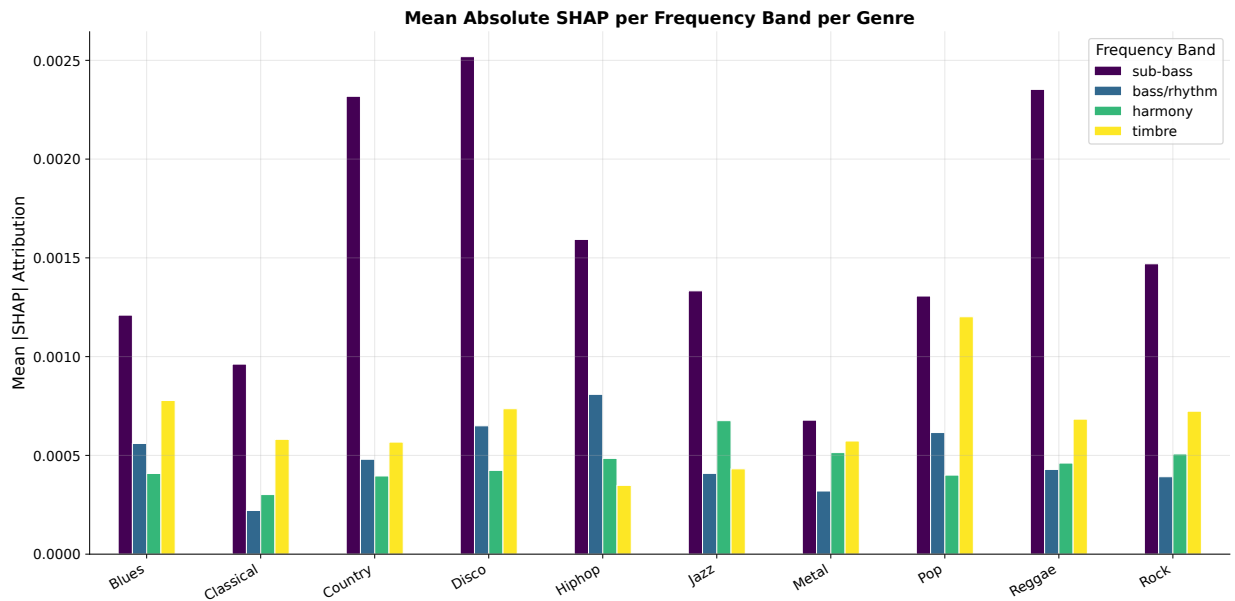


Figure 3. Mean absolute SHAP attribution per frequency band per genre, revealing genre-specific spectral attention patterns.

ranges from 6.0% (classical) to 18.8% (country).

This finding has significant implications. SHAP, operating at pixel granularity via gradient-based approximation, and LIME, operating at superpixel granularity via perturbation sampling, highlight fundamentally different aspects of the model’s decision process. A practitioner using only SHAP would arrive at a different explanatory narrative than one using only LIME, even for the same prediction. This confirms and extends the disagreement problem identified in the general XAI literature [21] to the MIR domain.

4.7 Spurious correlation detection

Table 5 and Figure 9 present the spurious correlation analysis, focusing on the silence attribution ratio.

Blues (ratio = 1.40) and **pop** (ratio = 1.20) show silence ratios exceeding 1.0, meaning the model assigns greater importance to spectrogram bins corresponding to silence

than to bins with actual musical content. This is a clear indicator of artifact exploitation. Blues tracks in GTZAN tend to be older recordings with distinctive noise floors; the model has evidently learned to recognise these recording-quality signatures rather than (or in addition to) the 12-bar blues structure and pentatonic scale patterns that define the genre musically. Similarly, pop tracks may carry characteristic compression artifacts that serve as shortcuts for the model.

For well-classified genres like hip-hop (ratio = 0.49), country (0.40), and jazz (0.51), the model appropriately focuses on high-energy musical content rather than silence regions.

Table 6 summarises all four XAI metrics jointly across genres.

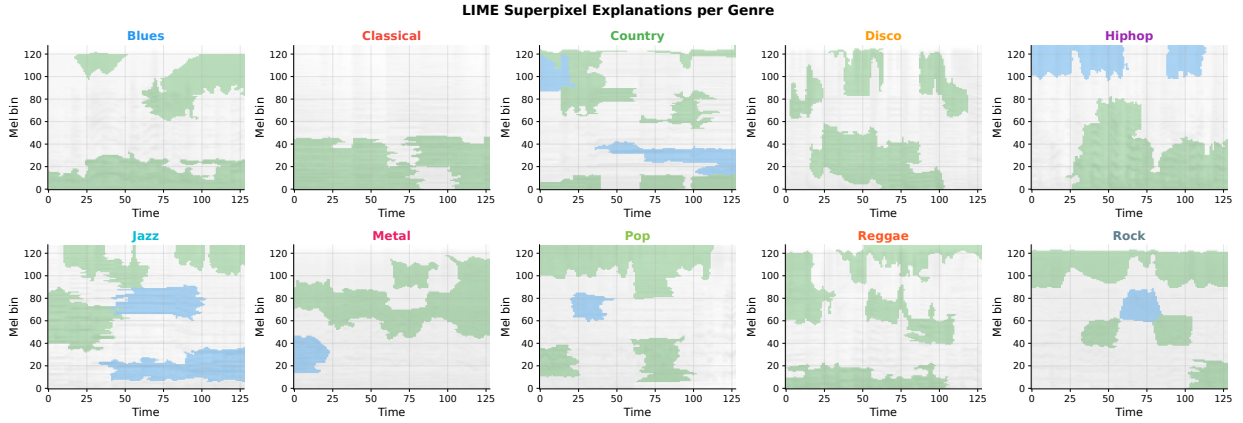


Figure 4. LIME superpixel importance maps for correctly classified clips. Green regions indicate positive contribution to the predicted genre; red indicates negative contribution.

Genre	ρ_{align}	p -value
Hip-hop	+0.949	0.051
Reggae	+0.200	0.800
Disco	0.000	1.000
Classical	-0.400	0.600
Jazz	-0.400	0.600
Metal	-0.400	0.600
Blues	-0.800	0.200
Rock	-0.800	0.200
Pop	-0.949	0.051
Country	-1.000	<0.001

Table 3. Musicological alignment: Spearman rank correlation between SHAP frequency-band attributions and expected musicological importance rankings. Positive values indicate alignment; negative values suggest artifact exploitation.

Genre	Pearson r	Spearman ρ	IoU ₁₀ (%)
Blues	0.037	0.039	11.4
Classical	0.038	-0.001	6.0
Country	0.165	0.145	18.8
Disco	-0.011	0.044	6.6
Hip-hop	0.173	0.208	9.7
Jazz	0.111	0.100	9.2
Metal	0.202	0.239	8.8
Pop	0.097	-0.004	13.8
Reggae	0.116	0.087	18.7
Rock	0.038	0.141	8.4

Table 4. Inter-method agreement between SHAP and LIME. Low correlations and IoU values indicate substantial disagreement across all genres.

4.8 Transformer attention analysis

Figure 10 displays CLS token attention heatmaps from the fine-tuned AST for representative clips of each genre. The attention patterns complement the SHAP and LIME findings: genres with concentrated attention over specific time-frequency regions (e.g., hip-hop’s low-frequency focus) generally correspond to higher musicological alignment scores, while genres with diffuse or edge-concentrated attention (e.g., blues) correspond to lower alignment.

5. DISCUSSION

5.1 A spectrum of musicological learning

The results reveal that genre classifiers do not uniformly learn genuine musical features or uniformly exploit artifacts. Instead, each genre occupies a position along a spectrum. At one end, hip-hop classification is well-grounded in musicological features: the model attends to the sub-bass and bass regions where the genre’s defining rhythmic and harmonic patterns reside. At the other end, country and pop classification appears driven primarily by artifacts,

with the model attending to frequency bands inversely correlated with musicological expectations.

This genre-dependent behaviour likely reflects the interaction between genre-specific acoustic properties and GTZAN’s dataset characteristics. Hip-hop’s spectral signature, dominated by low-frequency 808-style bass and drum machine patterns, is highly distinctive and consistent across recordings regardless of recording quality. In contrast, country and pop tracks in GTZAN may share less distinctive spectral signatures, leading the model to rely on secondary cues introduced by the dataset’s construction.

5.2 The disagreement problem in MIR

The finding that SHAP and LIME produce low-agreement explanations (Pearson $r = 0.04$ – 0.20) has practical implications for MIR researchers and practitioners. If one were to diagnose model behaviour using only SHAP, one would identify fine-grained spectral regions as important. Using only LIME, one would identify broader contiguous regions, potentially suggesting different musical features as salient. This disagreement arises from methodological differences: SHAP computes pixel-level Shapley values via gradient-based approximation, while LIME perturbs

Quantitative XAI Evaluation Results

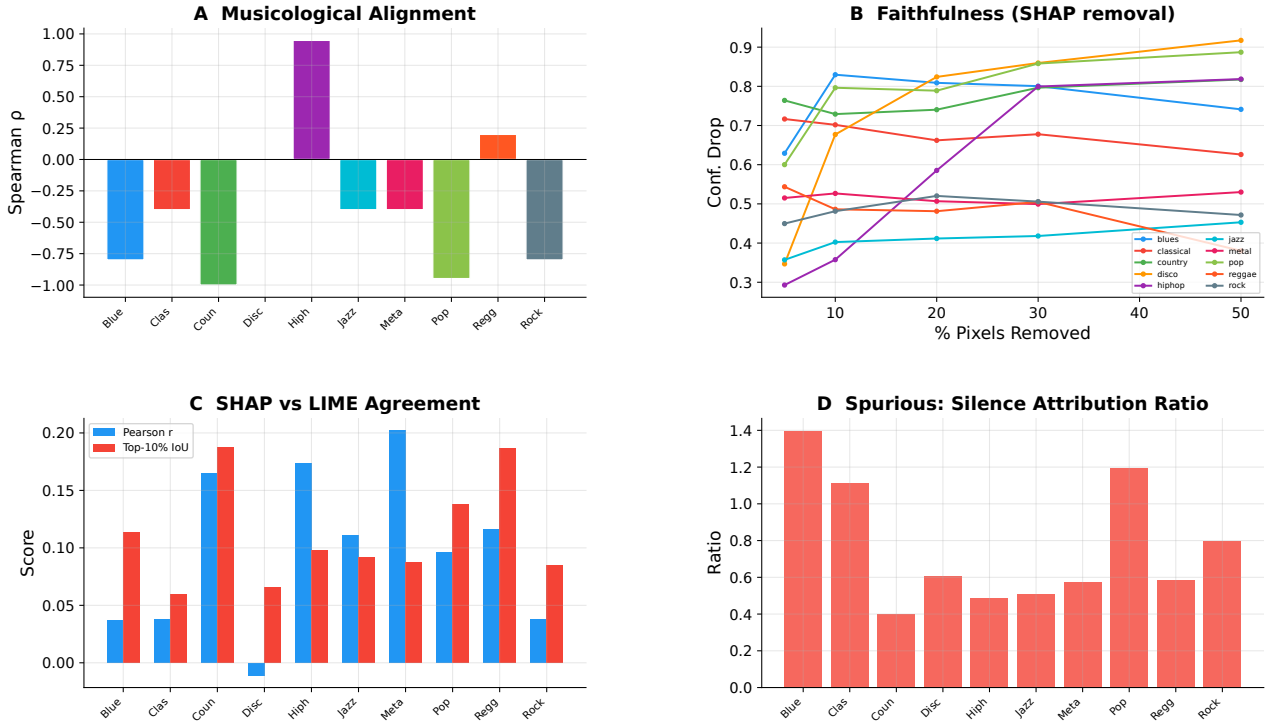


Figure 5. Quantitative XAI evaluation results. (A) Musicological alignment (Spearman ρ) per genre. (B) Faithfulness curves: prediction confidence drop under progressive SHAP-pixel masking. (C) SHAP vs. LIME agreement (Pearson r and top-10% IoU). (D) Spurious correlation: silence-to-high-energy SHAP attribution ratio.

Genre	Silence	High-E	Ratio	Flag
Blues	1.65e-3	1.18e-3	1.40	!
Pop	1.38e-3	1.16e-3	1.20	!
Classical	7.45e-4	6.69e-4	1.11	~
Rock	8.18e-4	1.02e-3	0.80	
Disco	1.09e-3	1.79e-3	0.61	
Reggae	9.75e-4	1.67e-3	0.59	
Metal	4.53e-4	7.89e-4	0.57	
Jazz	7.25e-4	1.43e-3	0.51	
Hip-hop	6.99e-4	1.44e-3	0.49	
Country	7.42e-4	1.87e-3	0.40	

Table 5. Spurious correlation detection: silence-to-high-energy SHAP ratio. Ratios exceeding 1.0 (!) indicate the model attributes more importance to silence than to musical content. Silence = mean |SHAP| in low-energy bins; High-E = mean |SHAP| in high-energy bins.

Table 6. XAI metrics summary per genre. Alignment ρ : Spearman rank correlation with musicological expectations. Faith. ($k=50\%$): confidence drop after masking 50% of top-SHAP pixels. SHAP-LIME r : Pearson correlation between the two attribution maps. Spurious ratio: silence-to-high-energy SHAP attribution ratio.

Genre	Alignment ρ	Faith. ($k=50\%$)	SHAP-LIME r	Spurious
Blues	-0.800	0.741	0.037	
Classical	-0.400	0.626	0.038	
Country	-1.000	0.817	0.165	
Disco	0.000	0.917	-0.011	
Hip-hop	+0.949	0.819	0.173	
Jazz	-0.400	0.453	0.111	
Metal	-0.400	0.530	0.202	
Pop	-0.949	0.887	0.097	
Reggae	+0.200	0.380	0.116	
Rock	-0.800	0.472	0.038	

superpixel-scale regions and fits a local linear model.

This disagreement is not a deficiency of either method alone, but rather a fundamental property of trying to explain complex, nonlinear decisions through simpler approximations. Each method captures a different facet of the model’s decision surface. This finding supports the principle of *multi-method triangulation*: explanations should be sought from diverse methods, and only conclusions supported by convergent evidence across methods should be

treated as robust.

5.3 Implications for dataset design

The spurious correlation analysis provides direct, quantitative evidence for concerns raised by Sturm [5, 10] about GTZAN. The model’s reliance on silence-region characteristics for blues and pop classification confirms that recording-quality artifacts serve as classification shortcuts.

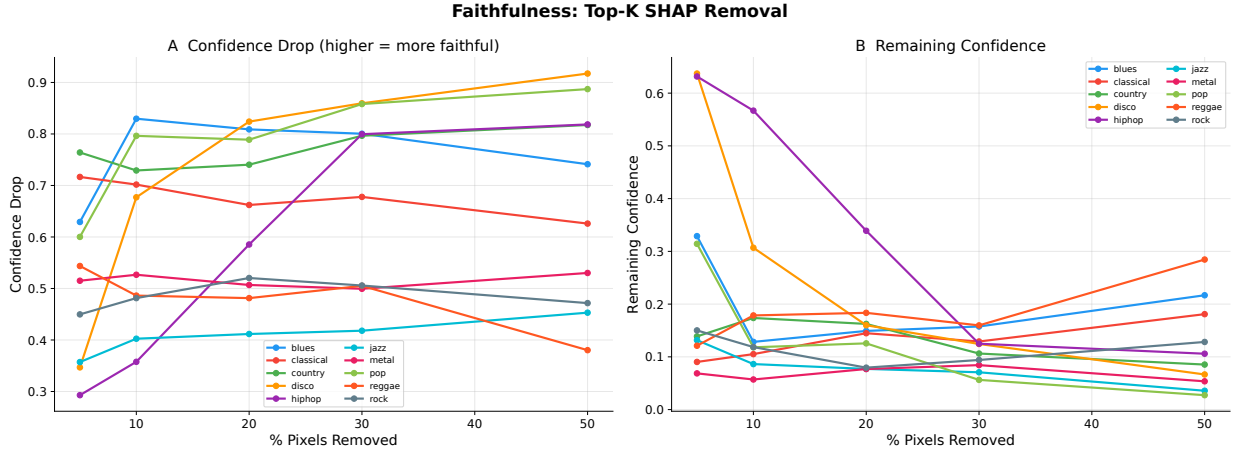


Figure 6. Faithfulness evaluation: prediction confidence drop as progressive fractions of the top-SHAP pixels are masked. Steeper curves indicate more faithful explanations.

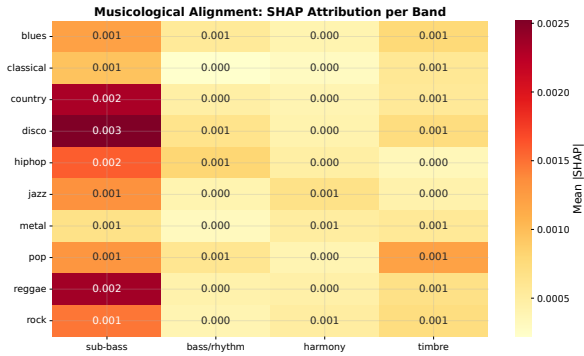


Figure 7. Musicological alignment scores by genre. Hip-hop shows strong positive alignment ($\rho = 0.95$) while country and pop show strong inverse alignment, suggesting artifact-driven classification.

This finding has implications beyond GTZAN: any music dataset with systematic genre-correlated recording conditions risks training models that exploit these correlations.

Future dataset construction should incorporate: (1) recording-quality normalisation across genres; (2) explicit artist-disjoint train/test splits; and (3) routine XAI-based auditing of trained models to detect spurious feature reliance before deployment.

5.4 Limitations

Several limitations should be acknowledged. First, the musicological ground truth adopted here is necessarily simplified: reducing genre-defining features to five frequency bands is a coarse approximation of the rich, multi-dimensional nature of musical genre. Second, the GTZAN dataset is small by modern standards (1,000 clips), limiting the statistical power of per-genre analyses. Third, the CNN architecture, while chosen for interpretability, does not represent the state of the art in genre classification; results may differ with more complex architectures. Fourth, the attention-as-explanation interpretation [18] remains debated in the literature, though this is mitigated

by using attention as a complementary signal alongside gradient-based and perturbation-based methods rather than as a standalone explanation.

6. CONCLUSION

This paper has presented a systematic, multi-method explainability analysis of a CNN music genre classifier on the GTZAN dataset, employing SHAP, LIME, and Transformer attention with a four-metric quantitative validation framework. The findings demonstrate that: (1) genre classifiers learn a genre-dependent mixture of genuine musicological features and dataset artifacts; (2) the choice of XAI method substantially affects the explanation narrative, with SHAP-LIME agreement consistently below $r = 0.21$; (3) spurious correlation detection can quantitatively identify genres where models exploit recording artifacts rather than musical content; and (4) multi-method triangulation provides a more complete and trustworthy picture than any single XAI method.

These results argue for adopting explainability analysis as standard practice in MIR model development: not only to understand *what* models learn, but to safeguard against deploying systems whose apparent competence masks reliance on dataset shortcomings.

7. REPRODUCIBILITY

All source code, trained models, and experimental outputs are available at [anonymised for review]. The pipeline consists of eight sequential scripts covering dataset preparation, CNN training, evaluation, SHAP/LIME computation, quantitative validation, AST fine-tuning, and figure generation. The GTZAN dataset is publicly available [1].

8. AI USAGE STATEMENT

Generative AI tools (GitHub Copilot) were used to assist with code implementation, debugging, and manuscript

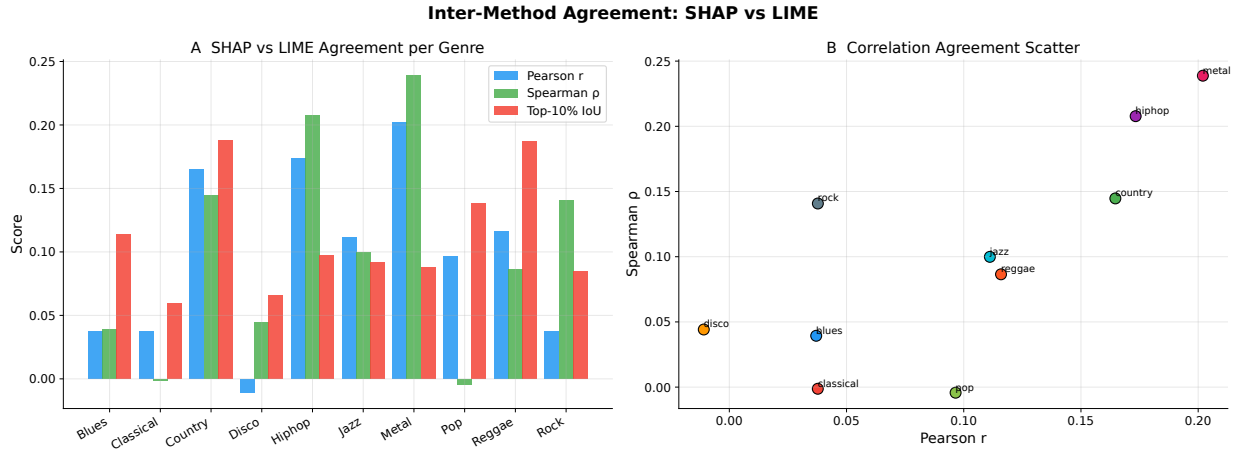


Figure 8. Inter-method agreement between SHAP and LIME. All three metrics (Pearson r , Spearman ρ , top-10% IoU) show consistently low agreement across genres.

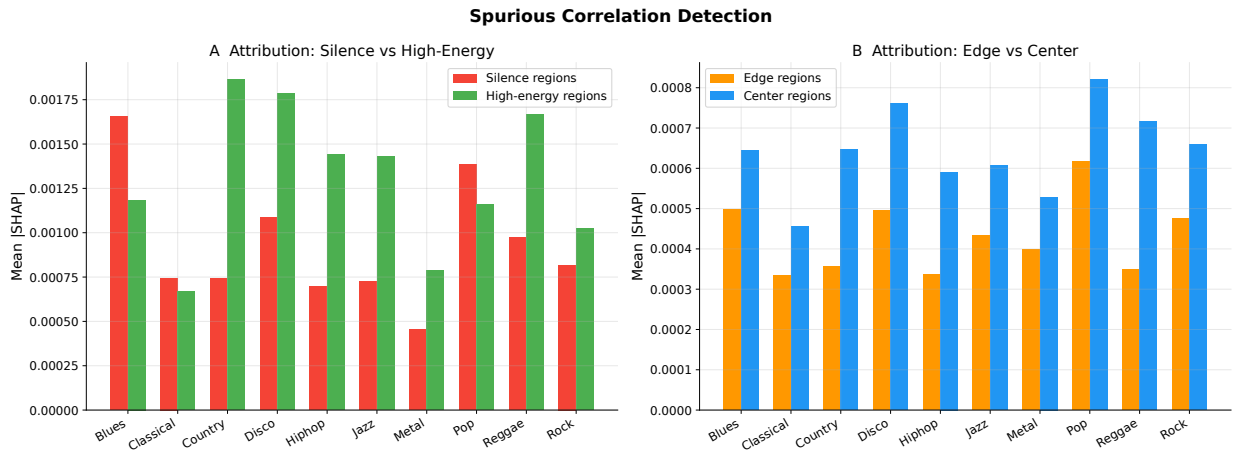


Figure 9. Spurious correlation analysis: SHAP attribution ratio between silence and high-energy spectrogram regions. Blues and pop show ratios exceeding 1.0, indicating artifact-driven classification.

drafting. All scientific content, experimental design, analysis, and conclusions were directed and verified by the author. The AI assistant did not independently design experiments or draw conclusions.

9. ETHICS STATEMENT

This research uses the publicly available GTZAN dataset [1], which contains short excerpts of commercially released music used under fair use for non-commercial research. No human subjects were involved.

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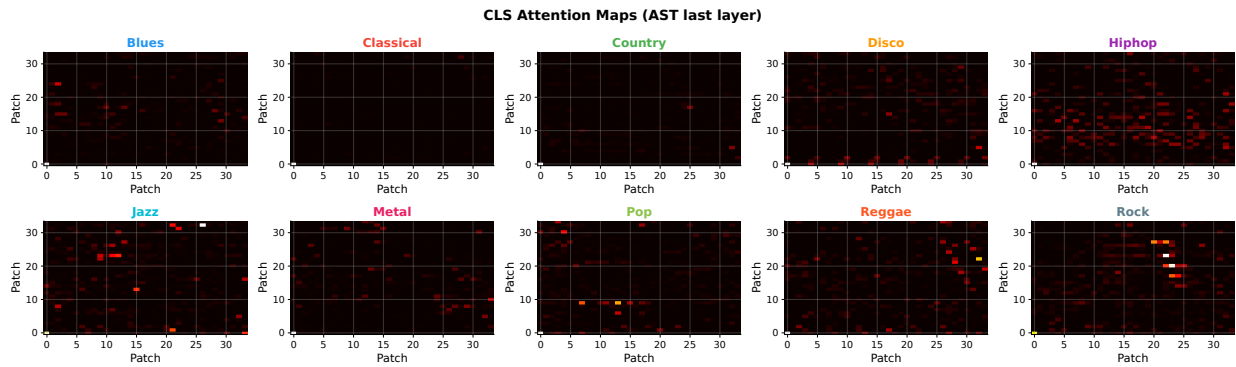


Figure 10. CLS token attention from the last self-attention layer of the fine-tuned AST, visualised over the input spectrogram for each genre. Brighter regions indicate higher attention weight.

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